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1.1.1. Calculate Momentum00:15

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 mass = float(input())
2 velocity = float(input())
3 momentum = mass * velocity
4 print(f"{momentum:.2f}kgm/s")
```

Average time0.006 s6.00 msMaximum time0.009 s9.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 13 ms

Expected output5.018.050.00kgm/sActual output5.018.050.00kgm/s

Test case 25 ms

TerminalTest cases

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1.1.2. Conditional Calculation Based on the Number of Digits01:20

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2 n = int(input())
3 if 1 <= n <= 999:
4     if 1 <= n <= 9:
5         print(n * n)
6     elif 10 <= n <= 99:
7         print(f"{math.sqrt(n):.2f}")
8     else:
9         print(f"{n ** (1/3):.2f}")
10 else:
11     print("Invalid")
```

Average time0.003 s3.14 msMaximum time0.004 s4.00 ms

4 out of 4 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 14 ms

Expected output981Actual output981

Test case 24 ms

Test case 34 ms

TerminalTest cases

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1.1.3. Days between Two Dates05:33

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2 date1 = input().strip()
3 date2 = input().strip()
4 date1 = datetime.strptime(date1, "%Y-%m-%d")
5 date2 = datetime.strptime(date2, "%Y-%m-%d")
6 difference = (date2 - date1).days
7 print(difference)
```

Average time0.008 s8.25 ms

Maximum time0.022 s22.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 102 ms

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 24 ms

TerminalTest cases

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1.1.4. Reverse a Number24:05

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2 rev = 0
3 while n > 0:
4     digit = n % 10
5     rev = rev * 10 + digit
6     n = n // 10
7 print(rev)
```

Average time0.004 s4.20 ms

Maximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 14 ms

Expected output

5367

7635

Actual output

5367

7635

Test case 25 ms

TerminalTest cases

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1.1.5. Multiplication Table

02:35

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 n = int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n*i}")
```

Average time

0.004 s

4.00 ms

Maximum time

0.005 s

5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

5 ms

Debug

Test cases

Expected output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Terminal

Test cases

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1.2.1. Pass or Fail

19:22

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorfa...

```
1 def calculate_grade(num_courses, marks):
2     if len(marks) != num_courses:
3         return "Error: Number of marks does not match number of courses."
4
5     if any(mark < 40 for mark in marks):
6         return "Fail"
7
8     aggregate_percentage = sum(marks)/num_courses
9
10    if aggregate_percentage > 75:
11        grade = "Distinction"
12    elif aggregate_percentage > 60:
13        grade = "First Division"
14    elif aggregate_percentage > 50:
15        grade = "Second Division"
16    elif aggregate_percentage >= 40:
17        grade = "Third Division"
18    else:
19        grade = "Fail"
20
21    return f"Aggregate Percentage: {aggregate_percentage:.2f}\nGrade: {grade}"
22
23
24 try:
25     num_courses = int(input().strip())
26     marks = list(map(int, input().strip().split()))
27     result = calculate_grade(num_courses, marks)
28     print(result)
29
30 except ValueError:
31     print("Error: Invalid input. Please enter integers only.")
```

Terminal

Test cases

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1.2.2. Fibonacci Series03:45

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     elif n == 2:
5         return 1
6     else:
7         return fibonacci(n - 1) + fibonacci(n - 2)
8
9
10 n = int(input())
11 for i in range(1, n + 1):
12     print(fibonacci(i), end=" ")
13
```

Average time0.011 s10.75 msMaximum time0.031 s31.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 13 ms

Expected output5

0 1 1 2 3

Actual output5

0 1 1 2 3

Test case 23 ms

TerminalTest cases

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1.2.3. Pattern - 102:15

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n = int(input())
2 for i in range(1, n + 1):
3     for j in range(i):
4         print("*", end=" ")
5     print()
```

Average time0.003 s3.00 msMaximum time0.004 s4.00 ms

2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 14 ms

Expected output

Actual output

TerminalTest cases

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1.2.4. Pattern - 210/20

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1n = int(input())
2for i in range(1,n+1):
3    for j in range(1,i+1):
4        print(j, end=" ")
5    print()
```

TerminalTest cases

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